Project Requirements Gathering Document

Project Track: Junior Data Analyst

Project Context: Smart OSH Performance & Risk Predictive Dashboard in Power Plants

3. Requirements Gathering

3.1 Stakeholder Analysis

This section identifies key stakeholders involved in the power plant safety ecosystem and outlines their core operational needs:

- **Safety Officers & OSH Team:** Need detailed, granular views of historical safety logs, risk classifications, and incident trends to perform root-cause analysis and deploy daily safety compliance protocols.
- **Plant Operations Managers:** Require cross-filtered operational metrics and unit-specific safety indicators to minimize downtime, monitor hazard patterns, and ensure overall plant compliance.
- **Executive Management / Decision Makers:** Require a high-level **OSH Executive Dashboard** summarizing macro safety trends, KPI baselines, and resource utilization to make informed budgeting and long-term safety policy decisions.
- **BI Project Team (Data Analysts):** Responsible for ensuring robust data integration, modeling the star schema, and validating DAX performance metrics to guarantee dashboard reliability.

3.2 User Stories & Use Cases

Practical user interactions with the dashboard to achieve specific occupational health and safety goals:

- **User Story 1 (Safety Officer):** *As a Safety Officer, I want to filter incident logs by hazard category and specific plant units so that I can immediately pinpoint areas where risk levels are escalating.*
- **User Story 2 (Plant Manager):** *As a Plant Manager, I want to monitor the update rates and active KPIs so that I can ensure operational data reflects current working conditions.*
- **User Story 3 (Executive Director):** *As an Executive Director, I want a single aggregated view of safety performance across all plants to measure user adoption and safety protocol effectiveness.*

Use Case: Risk Indicator Review and Drill-Through

Primary Actor: Safety Officer

Preconditions: Safety logs and environmental data have been cleaned and transformed using Power Query.

Main Success Scenario:

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1. The Safety Officer accesses the *Risk Classification Dashboard*.
2. The system dynamically runs DAX rules to classify records into **Low, Medium, or High** risk levels.
3. The dashboard flags an anomaly or a surge in 'High' risk alerts in a specific sector.
4. The Safety Officer applies a time filter and uses **drill-through capabilities** to inspect the underlying raw incident logs.
5. The Safety Officer generates a data-driven report to initiate a mitigation plan.

3.3 Functional Requirements

The core system capabilities and technical attributes that the Power BI solution must deliver:

- **FR-01: Data Ingestion & Cleansing:** The system must import multi-source records (Excel/CSV) and leverage Power Query to eliminate duplicates, resolve missing variables, and conform data types.
- **FR-02: Star Schema Modeling:** The platform must execute a robust data model establishing clear, direct relationships between the central incident fact table and corresponding dimension tables (Time, Plant Location, Severity, Hazards).
- **FR-03: DAX Performance Computations:** The dashboard must calculate live measures and calculated columns for key OSH performance metrics and rule-based risk triggers.
- **FR-04: Multi-Tiered Reporting Interfaces:** The application must present four standalone dashboard layouts: Executive view, Incident & Hazard analysis, Risk Classification overview, and Safety Performance comparisons.
- **FR-05: Dynamic Slicing & Filtering:** The system must allow users to interactively filter all visuals simultaneously by dates, plant sections, hazard categories, and severity tiers.

3.4 Non-Functional Requirements

The qualitative benchmarks and constraints to ensure system efficiency and user satisfaction:

- **NFR-01: Performance & Response Time:** Visual rendering, cross-filtering, and dynamic slicing queries must return results in under **2.5 seconds** during active navigation.
- **NFR-02: System Update & Data Freshness:** Scheduled data refreshes must run seamlessly without degrading visual performance, ensuring safety records stay aligned with operational reporting timelines.
- **NFR-03: Usability & Intuitive Design:** The workspace must utilize a clear layout featuring standardized, visual color coding (e.g., Red for High Risk, Yellow for Medium, Green for Low) accessible to non-technical operations personnel.
- **NFR-04: Reliability & Data Integrity:** DAX-calculated columns and rule-based parameters must yield mathematically accurate and consistent aggregates regardless of the user's selected filter combinations.
- **NFR-05: User Adoption Rate:** The reporting architecture must be designed intuitively to promote rapid onboarding, targeting a high **user adoption rate** across all plant supervising teams.

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